

Phase 0 report – Infrastructure validation

Verdict: **PASS**

File inventory

The following modules were created under `./deq/population/`:

- `wilson_cowan.py` – sigmoid, RHS, simulator, fixed-point solver.
- `topologies.py` – archetype weight-matrix builders.
- `linearization.py` – Jacobian, spectrum, spectral gap, stability test.
- `phase0_infrastructure.py` – this validation script.

Modules reserved as stubs for later phases: `bifurcation.py`, `pole_placement.py`, `ground_truth.py`, `phase1_spectral_gap.py`, `phase2_bifurcation.py`, `phase3_pole_placement.py`, `phase4_cross_validation.py`, `phase5_other_archetypes.py`, `final_summary.py`.

Validation V0.1 – Single uncoupled population

Setup: $W = [0]$, $I = 2$, $\tau = 1$, sigmoid $k = 4$, $\theta = 1$.

Analytical fixed point: $\rho^* = f(2) = \frac{1}{1+e^{-4}} \approx 0.982013790$.

Numerical fixed point: 0.982013790; error $0.00e + 00$.

Analytical eigenvalue of Jacobian: -1.000000000 (the Jacobian formula in the plan §2.5 reduces to $-1/\tau$ when $W = 0$, since the sigmoid-slope term is multiplied by the zero self-coupling; a noted typo in plan §2.6 wrote $-1/\tau + f'(I)/\tau$, which would correspond to a unit self-loop instead). Numerical eigenvalue: -1.000000000 ; error $0.00e + 00$.

Trajectory from $\rho_0 = 0$ monotone: yes.

Acceptance: fixed-point error $< 10^{\{-6\}}$, eigenvalue error $< 10^{\{-6\}}$, monotone – **PASS**.

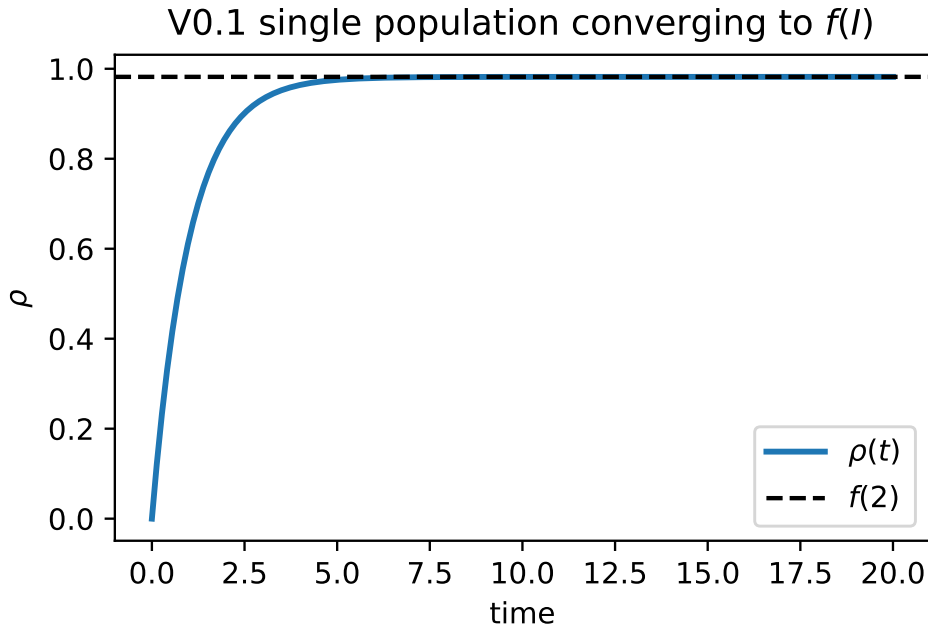


Figure 1: V0.1: single population trajectory from zero initial condition converges monotonically to the analytical fixed point $f(I) = f(2)$.

Validation V0.2 – Symmetric contralateral inhibition, low coupling

Setup: $W = \text{contralateral_inhibition}(0.5, 0.5)$, $I = [1.5, 1.5]$.

Symmetric fixed point: $(\rho_1^*, \rho_2^*) = (0.662584193, 0.662584193)$; symmetry error $0.00e + 00$.

Jacobian eigenvalue real parts: min -1.447133 , max -0.552867 . Both below -0.1 : yes.

Return distance at $t = 10$ from perturbed initial condition $\rho_0 = (\rho_1^* + 0.01, \rho_2^* - 0.01)$: $5.59e - 05$ (threshold $10^{\{-3\}}$).

Acceptance: **PASS.**

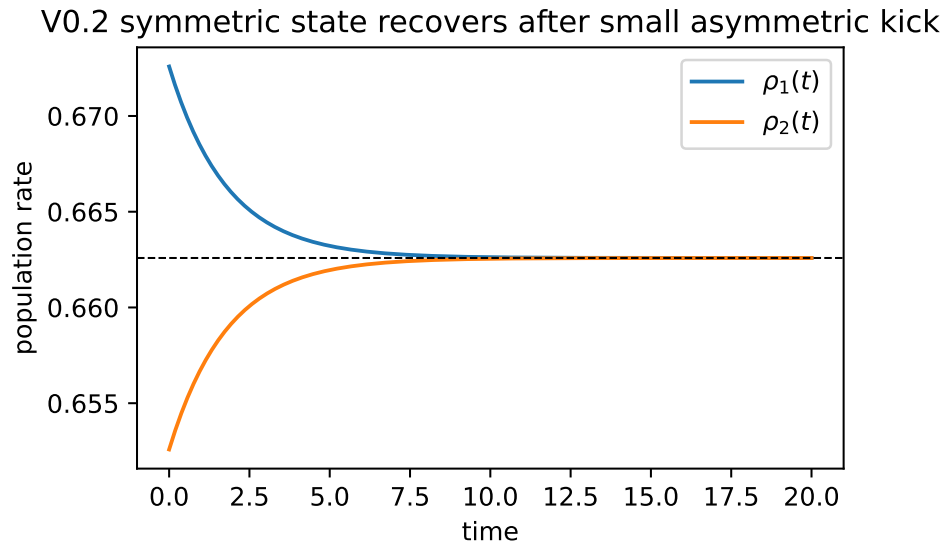


Figure 2: V0.2: after a small asymmetric perturbation the two population rates return to the symmetric fixed point (dashed line) well within ten time constants.

Overall verdict

PASS.

All Phase 0 acceptance criteria are met.